MAT 201 - MATHEMATICAL METHODS I

LECTURE 3: APPLICATIONS OF DIFFERENTIATION AND INTEGRATION

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OUTLINE

- Introduction
- Selected Applications
- Conclusion

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INTRODUCTION

Calculus, developed independently by Sir Isaac Newton and Gottfried Wilhelm Leibniz in the late 17th century, is a branch of mathematics that provides a framework for studying continuous change and motion through two fundamental concepts: differentiation and integration.

- Differentiation examines the instantaneous rate of change or slope of a function at a given point, revealing how quantities evolve in relation to each other.
- Integration deals with accumulating quantities over intervals, enabling the calculation of areas under curves and solving problems related to accumulation and summation.

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CALCULUS IS EVERYWHERE

Calculus is all around us. Whether in engineering, climate study, computing, or space travel, the practical uses of differentiation and integration are widespread.

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APPLICATIONS: COMPUTER MANUFACTURING

Newton's Law of Cooling predicts and manages electronic component temperatures in computer manufacturing. Engineers use it to model heat dissipation and design cooling systems, ensuring devices operate safely and efficiently.

$$\frac{dT}{dt} = -k(T - T_{\text{env}}) \tag{1}$$

where, T is the object's temperature, T_{env} is the surrounding environment's temperature, k is a positive constant, and $\frac{dT}{dt}$ is the rate of temperature change over time; the negative sign implies the object's temperature approaches that of the environment.

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APPLICATIONS: PREDICTING THE WEATHER

Calculus contributes to weather prediction by utilizing numerical models such as the primitive equations, a system of partial differential equations capturing atmospheric dynamics, with non-linear terms involving derivatives to simulate and forecast weather patterns, as exemplified by the quasi-geostrophic omega equation:

$$\frac{\partial(\nabla^2\omega)}{\partial t} + f\frac{\partial\omega}{\partial p} = -Q \tag{2}$$

where ω is the vertical velocity, f is the Coriolis parameter, p is pressure, and Q represents diabatic heating, capturing the role of calculus in weather prediction models.

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APPLICATIONS: EPIDEMICS ANALYSIS

Calculus is applied in epidemics analysis to model the spread of infectious diseases through differential equations. One such example is the SIR (Susceptible-Infectious-Recovered) model, a set of differential equations that describe the dynamics of an epidemic.

$$\left.\begin{aligned} \frac{dS}{dt} &= -\beta \cdot S \cdot I \\ \frac{dI}{dt} &= \beta \cdot S \cdot I - \gamma \cdot I \\ \frac{dR}{dt} &= \gamma \cdot I \end{aligned}\right\} \tag{3}$$

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APPLICATIONS: COSMOLOGY

Calculus is applied in cosmology to model the expansion of the universe, with the Friedmann Equations serving as fundamental differential equations describing the evolution of space-time geometry and matter content over cosmic time, represented by:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2} + \frac{\Lambda}{3} \tag{4}$$

where a represents the scale factor of the universe, ρ is the energy density, k is the spatial curvature constant, Λ is the cosmological constant, and G is the gravitational constant.

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APPLICATIONS: AUTOMOTIVE DESIGN

Calculus is applied in Automotive Design to optimize vehicle performance by modeling and analyzing dynamic aspects such as acceleration, deceleration, and suspension systems, with a sample equation being the calculus of variations used in minimizing fuel consumption by determining optimal trajectories.

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APPLICATIONS: BUILDING BRIDGES

Calculus is applied in building bridges to analyze structural forces, such as determining the maximum load-bearing capacity, and a sample equation could be the differential equation describing the bending of a beam,

$$EI\frac{d^2y}{dx^2} = -w(x) \tag{5}$$

where y is the deflection, x is the position along the beam, E is the modulus of elasticity, I is the moment of inertia, and $w(x)$ is the distributed load.

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APPLICATIONS: FINANCE AND BANKING

Calculus is applied in Finance and Banking for risk management through the computation of derivatives to assess financial instrument prices, with the Black-Scholes equation being a notable example in option pricing:

$$\frac{\partial C}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} + rS\frac{\partial C}{\partial S} - rC = 0 \tag{6}$$

where C is the option price, S is the asset price, r is the risk-free interest rate, σ is the volatility, and t is time.

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APPLICATIONS: ROCKETS AND SATELLITES

Calculus is essential in the field of Rockets and Satellites for trajectory analysis, with the differential equation of motion, such as the rocket equation

$$m\frac{dv}{dt} = -u\frac{dm}{dt} - mg + T, \tag{7}$$

where m is the mass, v is velocity, u is the exhaust velocity, T is thrust, and g is the acceleration due to gravity, playing a crucial role in predicting and optimizing their paths.

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APPLICATIONS: MICROWAVES

Calculus is applied in microwaves to analyze the rates of change in temperature during heating processes, and a sample equation illustrating this is the heat conduction equation:

$$\frac{\partial T}{\partial t} = \alpha \nabla^2 T, \tag{8}$$

where $\frac{\partial T}{\partial t}$ represents the rate of temperature change over time, α is the thermal diffusivity, and $\nabla^2 T$ denotes the Laplacian of temperature.

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APPLICATIONS: GAMBLING

Calculus is applied in gambling to determine optimal strategies and probabilities, with examples like the Kelly Criterion, expressed as

$$f^* = \frac{bp - q}{b}, \tag{9}$$

where f^* is the fraction of capital to bet, b is the odds received on the bet, p is the probability of winning, and q is the probability of losing.

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APPLICATIONS: DIGITAL MUSIC

In digital music, calculus is applied to model and optimize signal processing algorithms for tasks such as audio compression, where the discrete Fourier transform equation

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-i2\pi kn/N} \tag{10}$$

facilitates frequency analysis and data reduction techniques like MP3 encoding.

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CONCLUSION

The applications of calculus span a vast array of fields, from physics and engineering to economics and biology, demonstrating its indispensable role in understanding and modeling dynamic phenomena. Through differentiation and integration, calculus provides powerful tools for analyzing rates of change, finding accumulated quantities, and solving optimization problems, enabling advancements in technology, science, and everyday life. Whether it's predicting the motion of celestial bodies, optimizing financial investments, designing efficient structures, or decoding genetic sequences, calculus serves as a universal language for quantifying change and solving complex problems across diverse disciplines, underscoring its enduring relevance and impact on modern society.

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